

Marketing Acquisition Analysis

Overview

This project analyzes sampled marketing and user data to uncover trends and insights that can help Monarch optimize its acquisition strategy. The analysis focuses on channel efficiency, funnel performance, signup platform behavior, pricing and discounting effectiveness, and identifying anomalies that impact CAC, ROAS, and lifetime value.

Objectives

The primary goals of this analysis are to:

- Evaluate acquisition performance across marketing channels
 - Understand conversion behavior through the signup, trial, and subscription funnel
 - Identify how signup platform impacts user quality and conversion
 - Assess the effectiveness of pricing and discounting strategies
 - Surface anomalies and structural inefficiencies affecting CAC and ROAS
 - Provide actionable recommendations to improve acquisition efficiency and long-term value
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Data Sources

The analysis is based on five input datasets loaded into BigQuery:

- Users: User-level signup data
 - Attribution: Internal attribution channel and user-selected attribution
 - Trial: Free trial start and end timestamps
 - Subscription: Subscription start, plan period, pricing, and modeled lifetime value
 - Ad Spend: Daily spend by marketing channel
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Data Modeling

The analysis is built on a small set of fact tables designed using dbt-style modeling principles.

Core Tables

- Fct_user: User-level table acquisition funnel from signup through trial and subscription.
- Fct_channel_signup_cohort_day: Daily channel performance CAC, ROAS, LTV.
- Fct_pricing_analysis: Paid user pricing, discounting, and value
- fct_funnel_summary: Aggregated funnel metrics by channel and platform

All SQL used to create these tables is included in the sql/ directory.

Key Analyses

The analysis covers the following areas:

Channel Performance

- Volume vs efficiency tradeoffs
- CAC per paid subscriber
- ROAS based on lifetime value
- Identification of consistently value-negative channels

Funnel Analysis

- Signup to trial conversion
- Trial to paid conversion
- Conversion velocity and early funnel drop-offs

Signup Platform Insights

- Conversion differences between web, mobile web, iOS, and Android
- Impact of platform mix on channel performance
- Identification of platform-driven inefficiencies

Pricing and Discounting

- Full price vs discounted user value
- Discount depth tradeoffs between efficiency and absolute LTV
- Monthly vs yearly plan performance
- Pricing behavior by channel and platform

Anomalies and Validation

- Persistent value-negative spend

- Extreme CAC spikes
 - Attribution and data quality checks
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Tools and Technologies

- BigQuery for data storage and SQL execution
- SQL for modeling and analysis